

Elicitation Techniques

Chapter 1: Gathering Requirements

Chapter Objectives

In this chapter, we will position elicitation in the requirements process by:

- Identifying an elicitation planning process
- Understanding the characteristics of information
- Listing the types of information required for a project
- Identifying sources of requirements in a project
- Distinguishing how to elicit nonfunctional requirements

Gathering Requirements



Defining Elicitation

Elicitation Planning

Elicitation Environment

Eliciting for Categories of Requirements

Why We Gather Information

- We gather information first to define the problem and then to define what is necessary for the solution
 - Information is gathered from all levels of the organization
 - Information is gathered using various methods
- The primary method for gathering information is with a face-to-face interchange
- Elicitation pulls the information from a variety of sources
 - Interviews, documents, interfaces, surveys, etc
 - Often called “raw data”

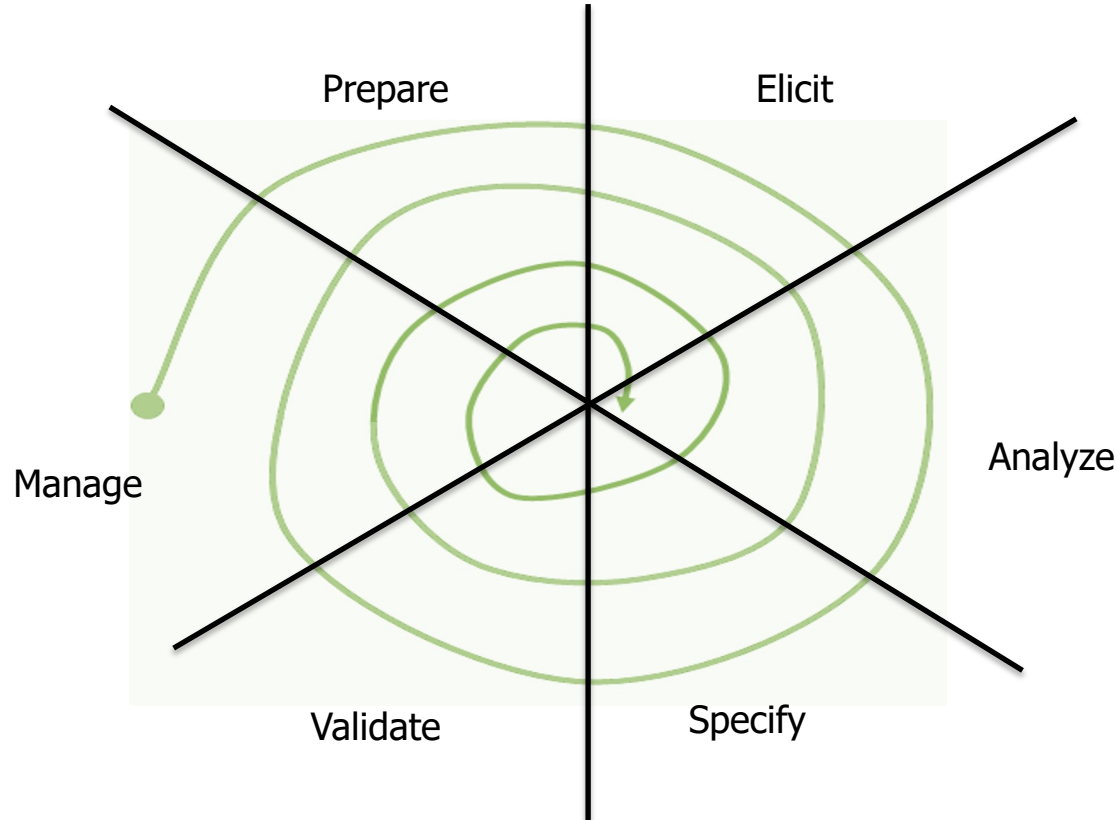


Connection Between SA and QA Roles



- Primary role of the SA is to gather this information
- QA role in elicitation is to:
 - Evaluate quality and integrity of data
 - Develop acceptance tests based on requirements
 - Seek additional information as required

The Requirements Spiral



What Is Elicitation?

- Elicitation centers on gathering and understanding what stakeholders really need based on:
 - What they say
 - What they do
 - What you see
 - What you discover
- Goal is to understand the:
 - Application domain
 - Users
 - Tasks users perform currently
 - Tasks users desire to perform as a result of the project
 - Environment in which the system will operate



What Is Information?



- Information = Raw data which has been structured or sequenced which provide facts about something
- Knowledge is gained by the analysis of information
- Sources of primary information
 - Information gathered specifically for the purpose
 - Result of interviews, surveys, observation
- Sources of secondary information
 - Not created for this purpose
 - Reports, policies and procedures manuals

Gathering Requirements

Defining Elicitation

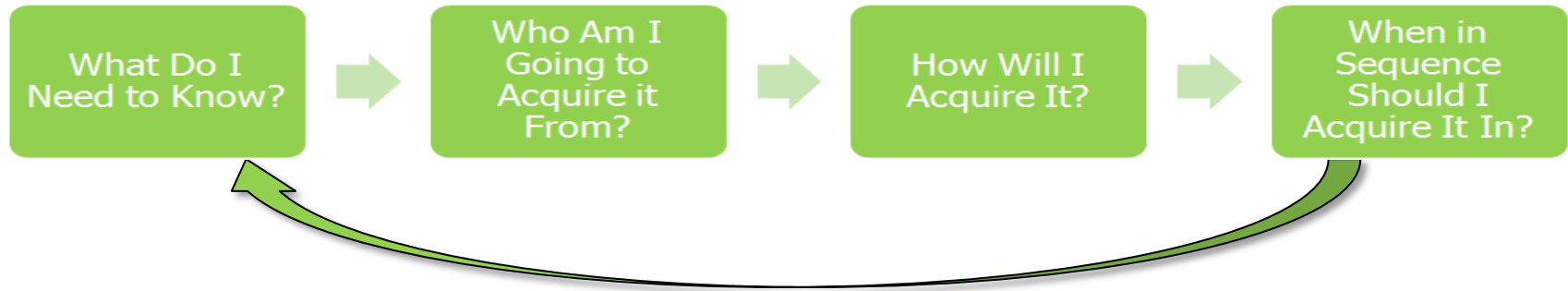


Elicitation Planning

Elicitation Environment

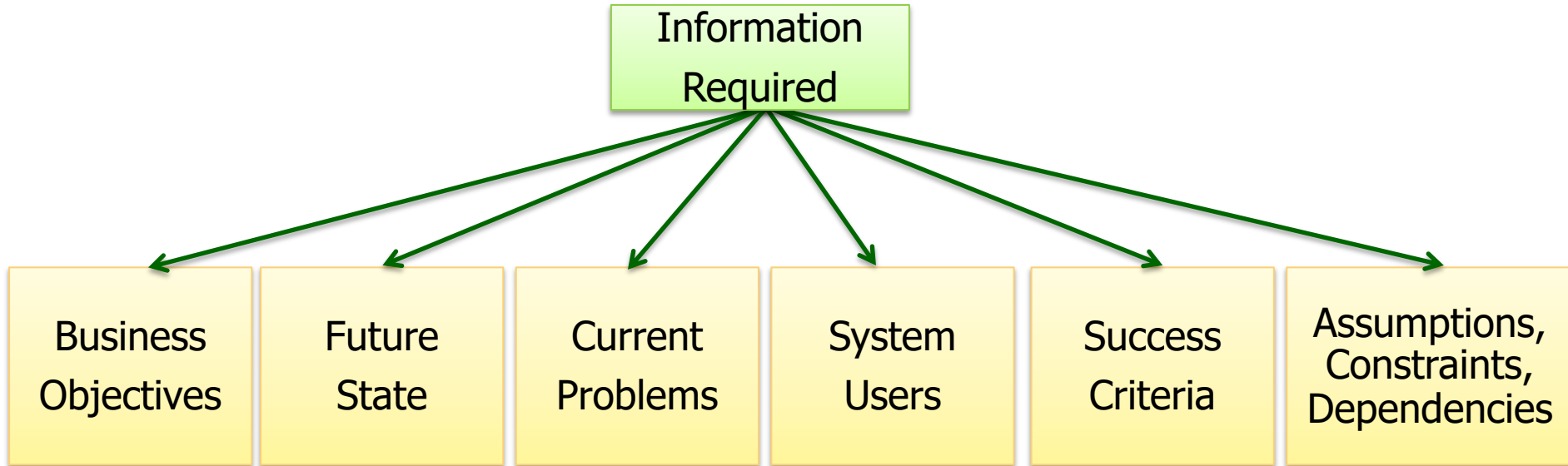
Eliciting for Categories of Requirements

The What-Who-How-When Elicitation Plan

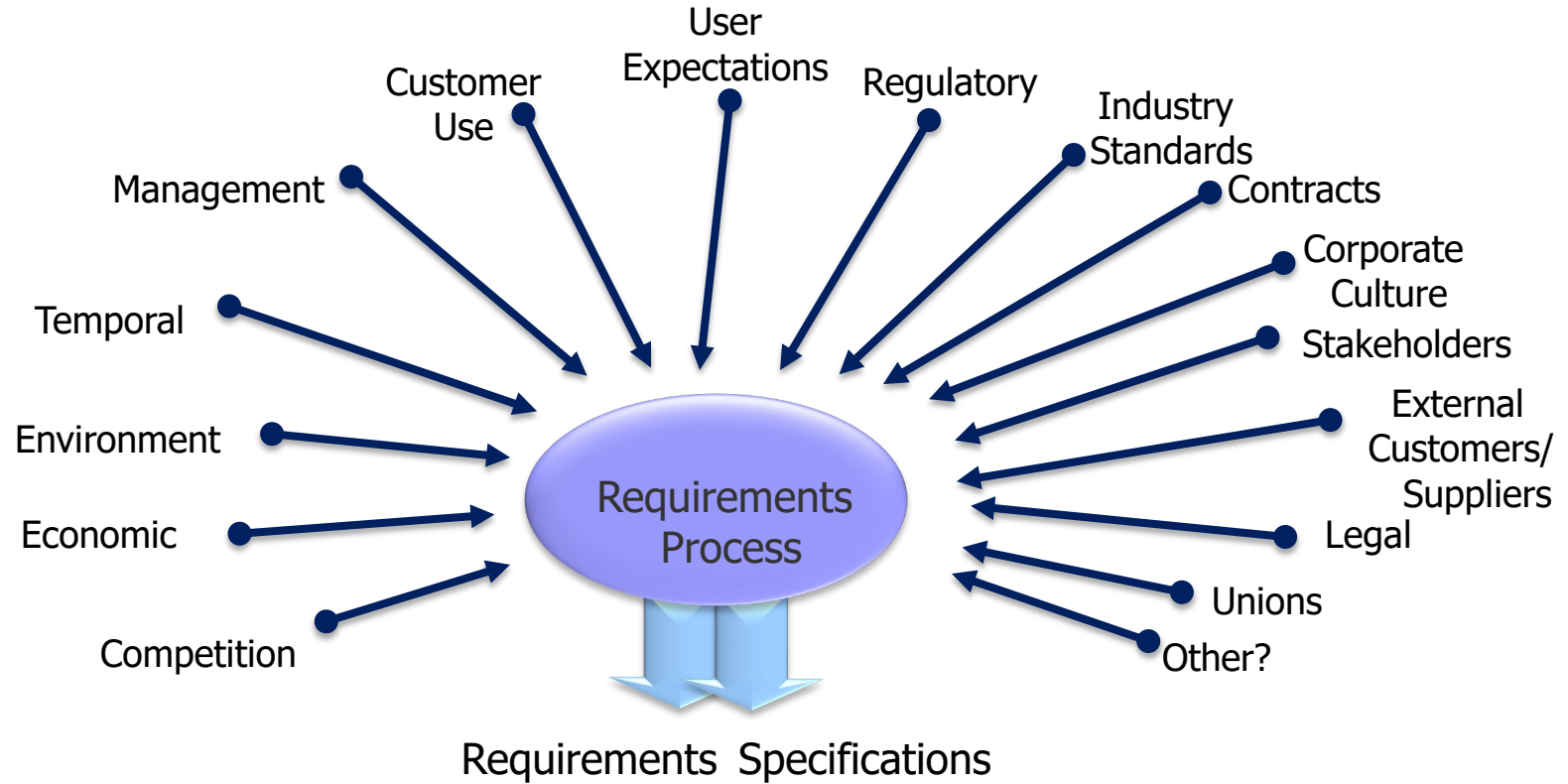


- Requirements elicitation requires a methodical, logical process of extracting product requirements from the customer and the ultimate operating environment
- The plan is to progressively elaborate from general to specific until enough details are known to guarantee that the product will solve the customer's problem

What Do I Need to Know?



Locating the Information (Sources of Requirements)



Gathering Requirements

Defining Elicitation

Elicitation Planning



Elicitation Environment

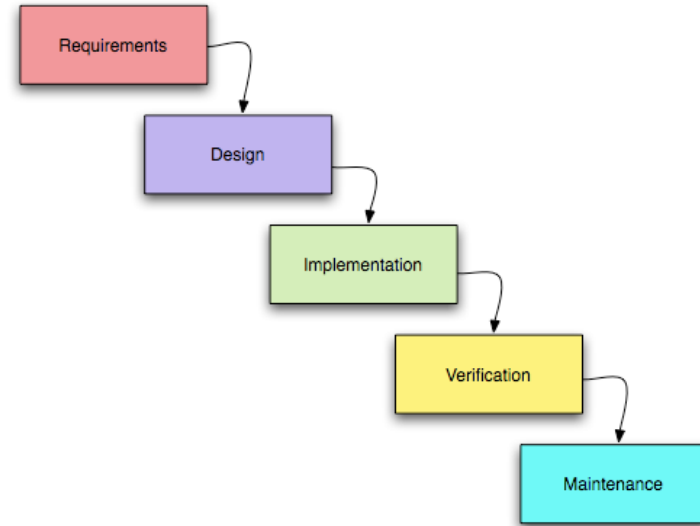
Eliciting for Categories of Requirements

Waterfall vs. Agile

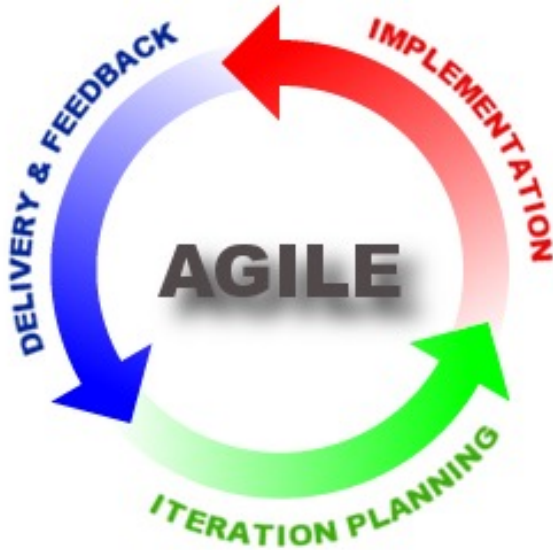
Waterfall	Agile
- Highly Structured	- Lightly Structured
- Requirements completed before design	- Requirements gathered Just In Time (JIT)
- Documentation critical	- Documentation optional (but required)
- Many stakeholders to interview	- Targeted stakeholders
- Many formal artifact (Process Map, Activity Diagram, etc.)	- No formal artifact, team creates what they require

Waterfall-based Elicitation

- Formal process
 - Requires an elicitation plan
 - Meetings must be well documented for transparency and accountability
 - Preparation is key to success
- Since the Analysis phase only happens once, everything must be collected
 - Comprehensive collection of requirements
 - Changing requirements after this phase will be costly
- Getting through gate to Design phase requires
 - All Analysis artifacts created and signed off
 - Test cases developed
 - Time/cost estimates for design phase



Agile-based Elicitation



- Ongoing developing process
 - Lots of interaction between analyst and user representative
 - User rep will be part of the development team
 - Much less need for formal interviews, we learn as we go
- Only need to collect requirements for a particular sprint cycle
 - Takes less time, easier to manage
 - May be further refined in future sprints
- Documentation needs decided by team members
- Focus is on acquiring just enough requirements to satisfy sprint's goals

Activity 1.2: Agile Elicitation Techniques



Goal	Discover how QA supports information gathering when using an Agile SDLC
Team Format	Teams of 4
Materials	Flip chart, laptops
Tasks	1. Brainstorm and research how elicitation might performed using Agile 2. Create a list of your top 5 techniques 3. For each technique, determine how QA might support the process 4. Create a flip chart which summarizes your findings 5. Prepare to defend your position
Deliverable	Flip chart of results
Duration	15 minutes

Gathering Requirements

Defining Elicitation

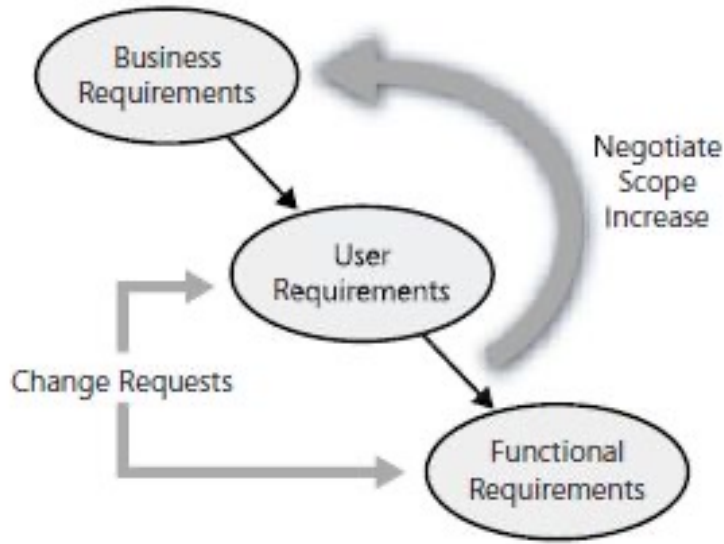
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Eliciting for Categories of Requirements

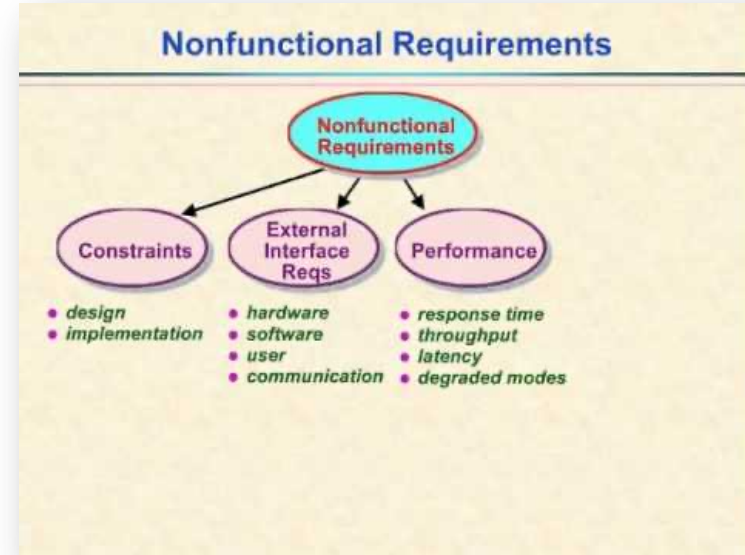
Functional Requirements



- Functional requirements state “what” the system must do
 - Elicitation centers around what the stakeholders want the system to do from a capability perspective
 - Stakeholders describe what they want
 - Defines the set of stakeholder expectations

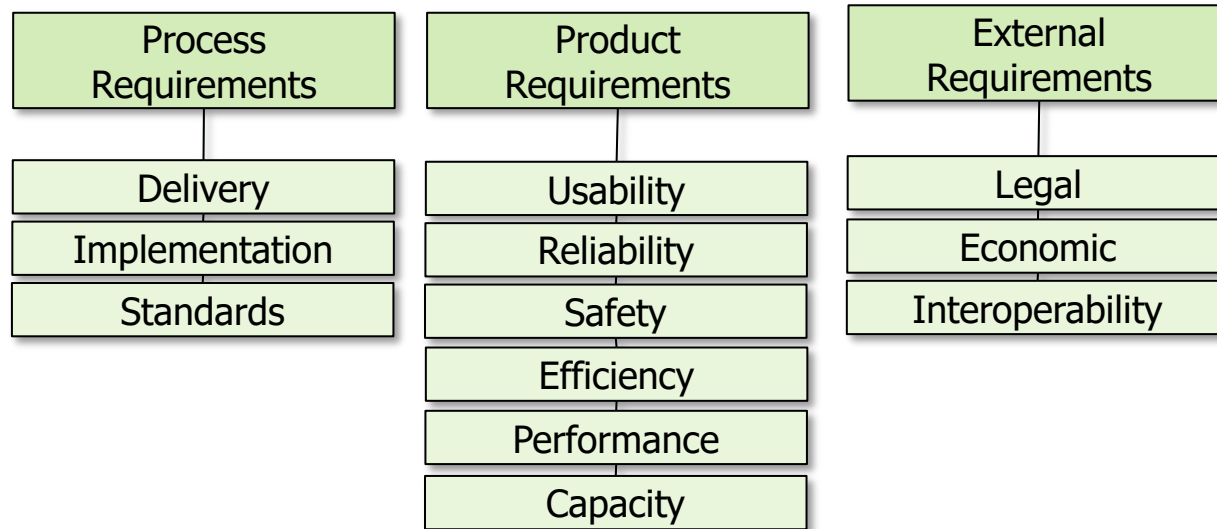
Nonfunctional Requirements

- Nonfunctional requirements define or constrain the environment in which the system will operate
 - Operating hours, performance characteristics, security, standards, etc
 - Generally closely tied to at least one functional requirement, but not always
 - Mostly intended for System Requirements (as opposed to User or Business)
- Nonfunctional requirements often stated by SMEs (QA, architects, network, security, etc.)
 - Sometimes proposed by the analyst and confirmed by the stakeholder



Areas of NonFunctional Requirements

- Guide stakeholders through their concerns using three main areas
 - Decompose into subareas
 - Continue to explore until no further concerns can be noted



ATM Example

■ Functional Requirements

- SR1.7 “The system shall allow the customer to withdraw money”

■ Associated Nonfunctional Requirements

- SR1.7.1 “Transactions should be completed within 5 minutes”
- SR1.7.2 “System shall respond to a request in less than 3 seconds”
- SR1.7.3 “Success is 90% of requests serviced in 3 seconds or less”
- SR1.7.4 “System must be secure when transmitting data”
- SR1.7.5 “Transmissions shall be encrypted”

■ The stakeholders will likely clearly describe the functional requirement

- ## ■ It is less likely that the stakeholders will identify any of the nonfunctional requirements
- The analyst must “discover” these requirements through elicitation
 - Note: Sometimes the stakeholder may not even be able to properly articulate their needs in this area

Activity 1.3: QA Analyst as Nonfunctional Requirements Gatherer



Goal	To identify questions an analyst asks for NFs
Team Format	Individually
Materials	This slide and previous
Tasks	- For the ATM example described earlier, identify 3 – 5 questions an QA analyst should ask a stakeholder in order to discover the nonfunctional requirements. _____ _____ _____ _____ _____
Deliverable	Be prepared to share your questions with the class
Duration	20 minutes

Chapter Summary

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